

f indexes firms, q indexes Eloundou et al. (2024) exposure quintiles, and t indexes months, with $t = -1$ corresponding to October 2022. The outcome variable $y_{f,q,t}$ is employment in f, q, t . Equation 4.1 is a Poisson event study regression controlling for firm-quintile effects, $\alpha_{f,q}$, and firm-time effects, $\beta_{f,t}$. The firm-time effects absorb aggregate firm shocks that impact each exposure quintile equally. The firm-quintile effects adjust for baseline differences in hiring across quintiles within the firm. The coefficients of interest, $\gamma_{q,t}$, measure differential changes in employment growth across quintiles after accounting for firm-time effects and firm-quintile effects.¹⁸

We run this regression separately for each age group. For each regression, we restrict to firms that hire at least 10 workers within the age group in every period of the sample. Further, $\sum_t y_{f,q,t}$ must equal at least 100 for each q , meaning that the firm must at least employ on average about 2 workers from each exposure quintile across months in the sample.¹⁹ Standard errors are clustered by firm.

Results are in Figure 4, which plots the $\gamma_{q,t}$ coefficients for each age group. For workers aged 22-25, estimates for higher quintiles are large and statistically significant, with a 15 log point decline in relative employment comparable in magnitude to the estimates in the raw data in Figure 2. Estimates for other age groups are generally much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

One alternative confounder that would not be controlled for with firm-time effects is that even conditional on the firm workers with high AI exposure were excessively hired after the COVID-19 pandemic, leading to a subsequent contraction in their hiring. To assess such alternatives we consider various other robustness checks in Section 4.6, such as removing computer occupations and conditioning on whether the occupation is amenable to work from home.

¹⁸Because of zero counts in the outcome variable, we estimate a Poisson regression instead of an OLS regression in logs following guidance from Chen and Roth (2024).

¹⁹Results are not sensitive to these restrictions, though there must be at least one non-zero value in each firm-month and each firm-quintile for observations to not get dropped in the Poisson regression.

occupation cells. Future work should study employment trends in other large-scale data sources such as the American Community Survey (ACS) or data from Revelio Labs as in [Hosseini and Lichtinger \(2025\)](#) and [Klein Teeselink \(2025\)](#).

5 Conclusion

We document six facts about the recent labor market effects of artificial intelligence.

- We find substantial employment declines for early-career workers in occupations most exposed to AI, such as software development and customer support.
- While economy-wide employment continues to grow, employment growth for young workers has been stagnant.
- Entry-level employment has declined in applications of AI that *automate* work, with muted effects for those that *augment* it.
- After conditioning on firm-time effects, young workers experienced a 16% relative employment decline in the most exposed occupations.
- Labor market adjustments are more visible in employment than in compensation.
- These patterns hold across various alternative analyses.

While our main estimates may be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment significantly.

The adoption of new technologies typically leads to heterogeneous effects across workers, resulting in adjustment periods as workers reallocate from displaced forms of work to new forms with growing labor demand ([Autor et al., 2024](#)). Past transitions such as the IT revolution ultimately led to robust growth in employment and real wages following physical and human capital adjustments, with some workers benefiting more than others ([Bresnahan et al., 2002](#); [Brynjolfsson et al., 2021](#)).

Tracking employment trends on an ongoing basis will help determine if the adjustment to AI follows a similar pattern. We will continue monitoring these outcomes to assess whether the trends